

1

2

3

4

5

6

A

B

C

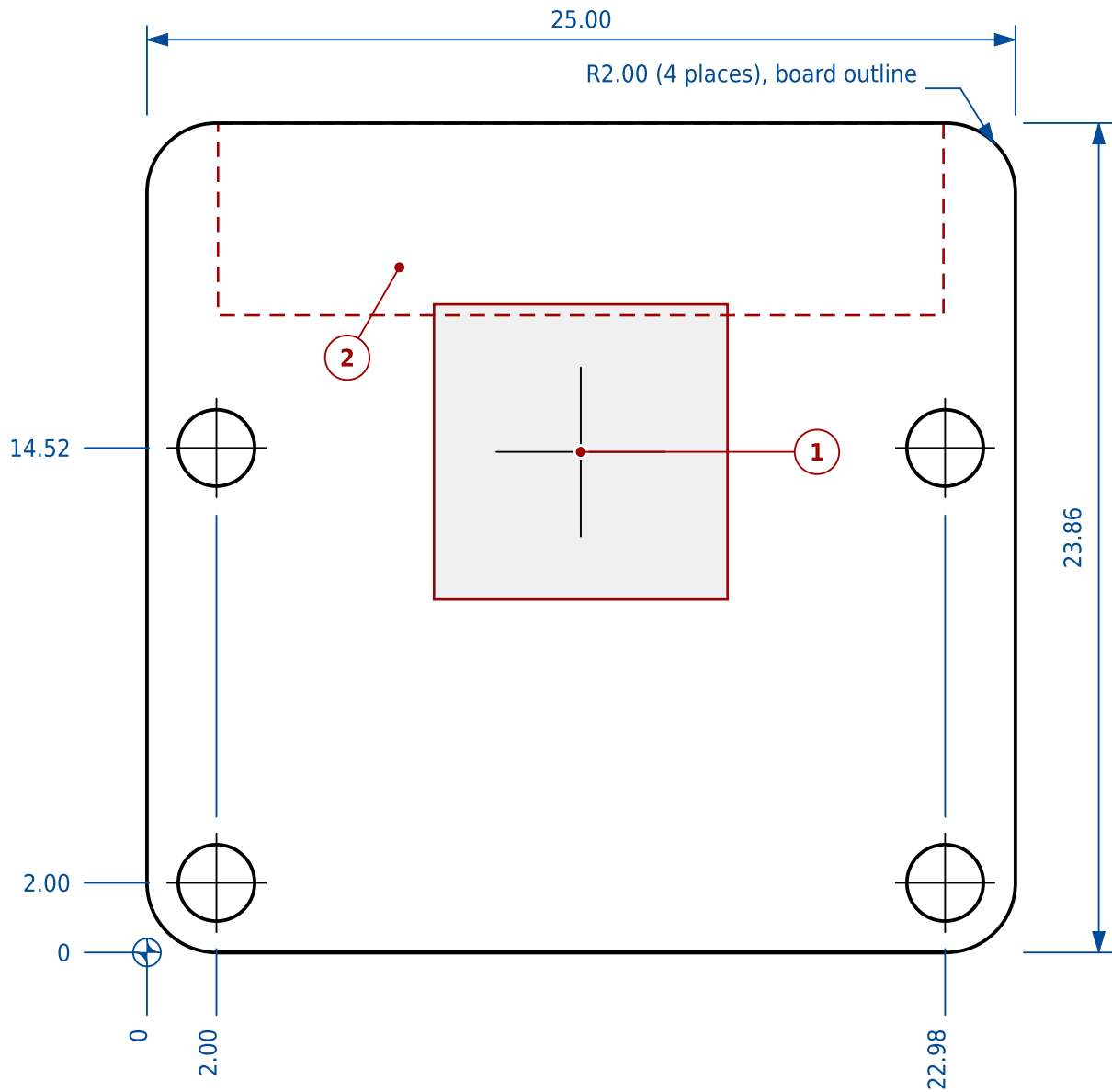
D

A

B

C

D



HOLE SCHEDULE

ID	X mm	Y mm	DIA mm	KEEPOUT mm
MT1	2.00	2.00	2.20	not given
MT2	22.98	2.00	2.20	not given
MT3	2.00	14.52	2.20	not given
MT4	22.98	14.52	2.20	not given

FEATURE SCHEDULE

#	FEATURE	X EXTENT mm	Y EXTENT mm
1	Lens and sensor module	8.26 to 16.71	10.16 to 18.65
2	Camera FFC connector, 15-way	2.05 to 22.93	18.33 to 23.85

LEGEND

- Board outline
- Component body, scheduled
- On the underside, seen through the board
- Dimension, extension and leader

NOTES

- Viewed from the component side.
- Hole IDs are this drawing's, the same hole on every camera sheet. The two drawings are read separately and their hole patterns agree to 0.025, which is why one sheet prints 22.98 where the other prints 23.00.
- Optical axis at X 12.49, Y 14.40, the centre of feature 1. Feature 1 is dimensioned 8.5 square. Its lower body, stepped, reaches to 2.76 from the lower edge across 6.72 to 16.14.
- The source drawing is a plan view only and gives no height.
- Source read as a 1.5055:1 plot: scale from the 21.0 x 12.5 hole rectangle, overall dimensions then back within 0.031.
- Every figure printed on the source is an outlined path, not text: its geometry is machine-read, its printed dimensions are transcribed by eye.
- The component side is the lens side; the connector is on the far face.

SOURCES

- Mechanical drawing: <https://datasheets.raspberrypi.com/camera/camera-module-2-mechanical-drawing.pdf> - Raspberry Pi Ltd, RPI-CAM-V2_1, 12/11/2015
- Mounting holes: <https://datasheets.raspberrypi.com/camera/camera-module-2-mechanical-drawing.pdf> - Drawing note, quoted: 4x 2.2mm diameter holes. No land or keep-out is shown.

RASPBERRY PI CAMERA MODULE 2

25 x 23.862 mm, Sony IMX219

DRAWN generated		VERSION v0.0-317-gcd1696b		UNITS mm		SCALE 5:1	
SIZE A3	SHEET 1 OF 1		DRAWING NO RPICAM-2				REV A
MATERIAL PCB, thickness not stated			GENERAL TOLERANCE edge +/-0.20 hole pos +/-0.10 hole dia +/-0.08				

1

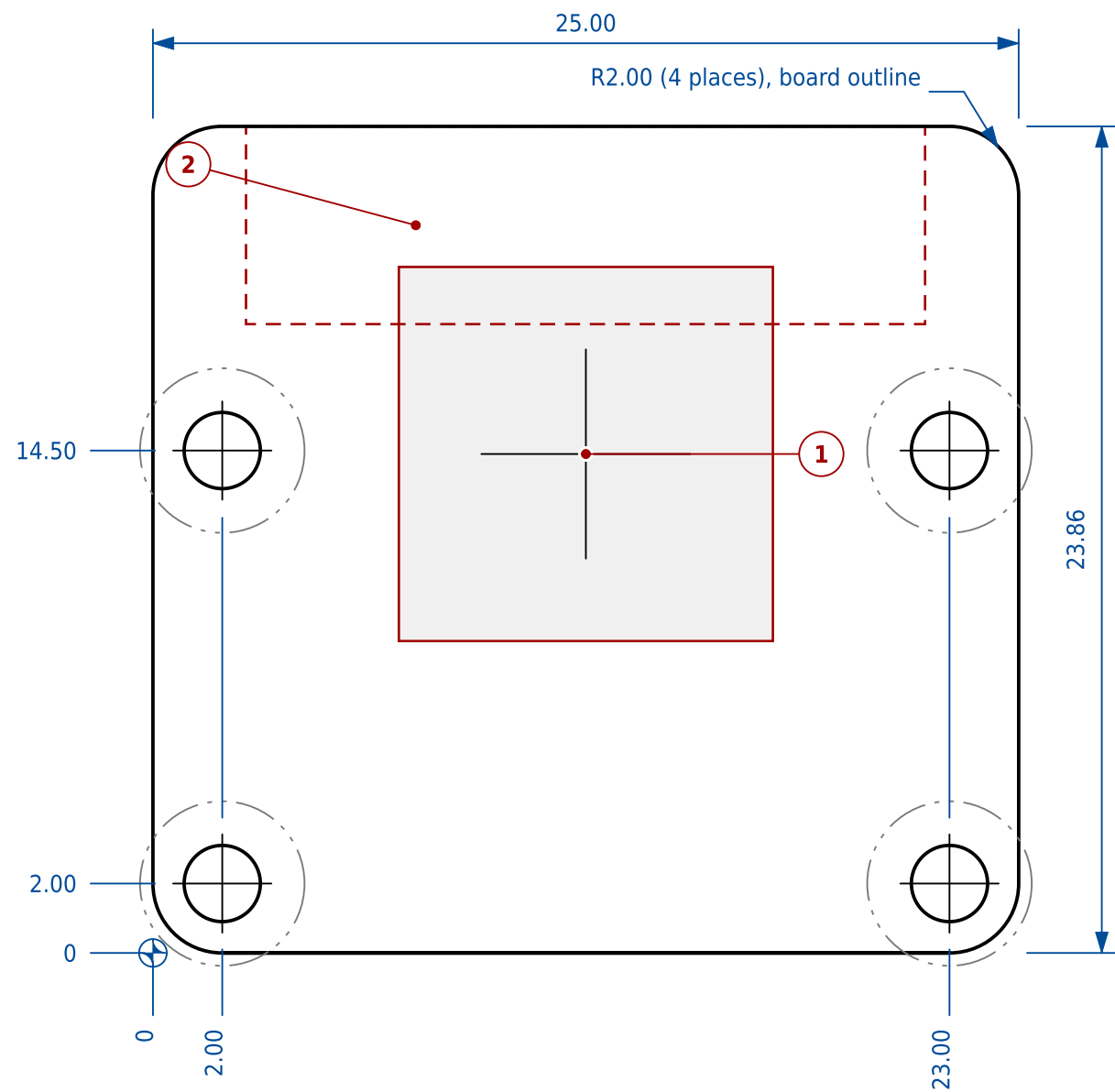
2

3

4

5

6



HOLE SCHEDULE

ID	X mm	Y mm	DIA mm	KEEPOUT mm
MT1	2.00	2.00	2.20	4.75
MT2	23.00	2.00	2.20	4.75
MT3	2.00	14.50	2.20	4.75
MT4	23.00	14.50	2.20	4.75

FEATURE SCHEDULE

#	FEATURE	X EXTENT mm	Y EXTENT mm
1	Lens and sensor module	7.10 to 17.90	9.00 to 19.80
2	Camera FFC connector, 15-way	2.69 to 22.30	18.15 to 23.86

LEGEND

- Board outline
- Component body, scheduled
- On the underside, seen through the board
- Hole keep-out
- Dimension, extension and leader

NOTES

- Viewed from the component side.
- Hole IDs are this drawing's, the same hole on every camera sheet. The two drawings are read separately and their hole patterns agree to 0.025, which is why one sheet prints 22.98 where the other prints 23.00.
- Optical axis at X 12.50, Y 14.40, the centre of feature 1. Clear aperture printed ø5.75 standard, ø6.95 wide; the wide drawing draws ø7.00. Feature 1's lower body, 8.9 across, reaches to 1.70 from the lower edge.
- Overall thickness printed 11.3 standard, 12 wide, lens tip to connector back; lens assembly 6.98 and 8.3 above the board. The source's own elevations scale 1.2 short, so no height is drawn here.
- Source read as a true 1:1 plot: scale from the 21.0 x 12.5 hole rectangle, overall dimensions then back within 0.004.
- The standard and wide drawings agree on the outline, holes and lens module within 0.05, so one sheet covers both.

NOTES (continued)

- The component side is the lens side; the connector is on the far face.
- MATERIAL gives the thickness the source states, 1.120 mm, which is within 0.05 mm of no standard finished thickness.

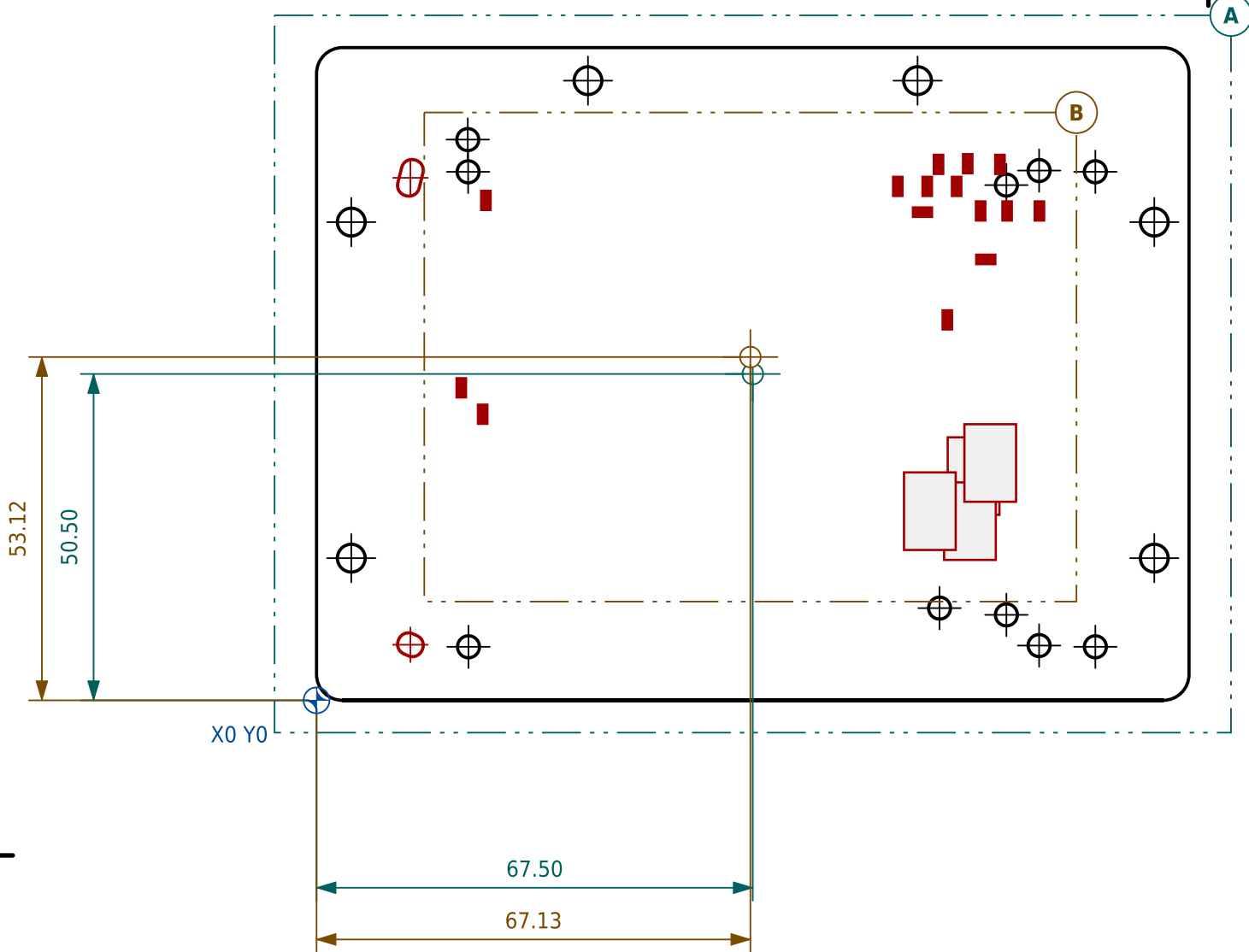
SOURCES

- Mechanical drawing: <https://datasheets.raspberrypi.com/camera/camera-module-3-standard-mechanical-drawing.pdf> - Raspberry Pi Ltd, RP-008153-DS-1
- Mechanical drawing: <https://datasheets.raspberrypi.com/camera/camera-module-3-wide-mechanical-drawing.pdf> - Raspberry Pi Ltd, RP-008155-DS-1
- Mounting holes: <https://datasheets.raspberrypi.com/camera/camera-module-3-standard-mechanical-drawing.pdf> - ø2.2, with an ø4.75 land round each one.

RASPBERRY PI CAMERA MODULE 3

25 x 23.862 mm, standard and wide, Sony IMX708

DRAWN generated		VERSION v0.0-317-gcd1696b	UNITS mm	SCALE 5:1	
SIZE A3	SHEET 1 OF 1	DRAWING NO RPICAM-3			REV A
MATERIAL PCB, 1.120 as given		GENERAL TOLERANCE edge +/-0.20 hole pos +/-0.10 hole dia +/-0.08			



NOTES

1. The camera looks straight down. X and Y locate the optical axis in the subject's plan frame; Z is the lens height above the plane its target lies in. ASSUMED: Z is to the entrance pupil, which no vendor locates; set it from the lens face.
2. A frame is the smallest rectangle of the sensor's own 4:3 holding its target plus 5.00 mm all round; LONG says which way the camera is turned. It is the shape of the picture, not of the optics, so both lenses share it and differ only in Z.
3. PLANE, which Z is measured from: A from the plate's own top face; B from the DEMO BOARD's top face, not the plate's -- the standoff height plus the board thickness, 1.56 to 1.60 mm, above the plate. The standoff height is the builder's and is specified nowhere here, so measure the stack. A stand set h mm too low covers only (Z - h) / Z of what is drawn, so the error always loses the edges.
4. HEADROOM: The demo board itself, any revision lies inside frame A in plan but stands above the plane its height is set from, and stays in the picture only up to 25.2 mm at 65 deg, 9.5 mm at 120 deg. Above that, raise the camera.

NOTES (continued)

5. FOCUS: Raspberry Pi give the depth of field as "Approx 1 m to infinity", so 1000 mm is the nearest this lens focuses. Every height here is nearer than that, so a stock OV5647 cannot focus on any of these boards. Use one whose lens focuses closer.
6. AUTOFOCUS: an autofocus OV5647 is a different part -- Arducam's B0176, declared 54 x 44 deg against the stock lens's 53.50 x 41.41, with a motorised lens and a close focus setting. Arducam publish no near limit and no lens height for it. Raspberry Pi's focusable modules, other sensors, are given as "Approx 10 cm to infinity" and "Approx 5 cm to infinity".
7. The 120 deg lens's declared pair is not self-consistent: 120 across a 4:3 sensor implies 104.82 down it, not 90. Z is the greater of what the two ask for, so the picture runs OVER the rectangle -- A by 44.26 mm in X, B by 30.18 mm in X -- which is still covered.
8. The "65 deg" name is the stock lens's diagonal, which no vendor prints. DERIVED: $2 \times \text{atan}(\sqrt{3.76^2 + 2.74^2}) / 2 / 3.60 = 65.74$ deg from the image area, 64.42 from the pixel array.

FRAMES, AND Z ABOVE THE FRAME PLANE, mm

	FRAME	RECTANGLE mm	LONG	AXIS X	AXIS Y	Z 65	Z 120
A	Whole plate	148.00 x 111.00	X	67.50	50.50	146.8	55.5
B	Every LED and 7-seg	100.91 x 75.69	X	67.13	53.12	100.1	37.8

LENSES

LENS	H deg	V deg	FOCUS	NEAR mm
65 deg (stock lens)	53.50	41.41	Fixed	1000
120 deg (wide lens)	120.00	90.00	Fixed	not published
Autofocus	54.00	44.00	Auto	not published

LEGEND

- Subject outline
- LED, display or connector
- - - Frame A, the rectangle the picture must cover
- - - Frame B, the rectangle the picture must cover
- ➡ Dimension, extension and leader

NOTES (continued)

9. Quotes here are transliterated to ASCII: x for the multiplication sign, um for micro, infinity for its symbol. The cached pages carry them as printed.
10. Frame B is the union of every LED and 7-segment over all five revision families; no one revision needs all of it. They are not clustered -- 90.91 x 62.79 of a 135 x 101 plate -- so frame B buys little over frame A.

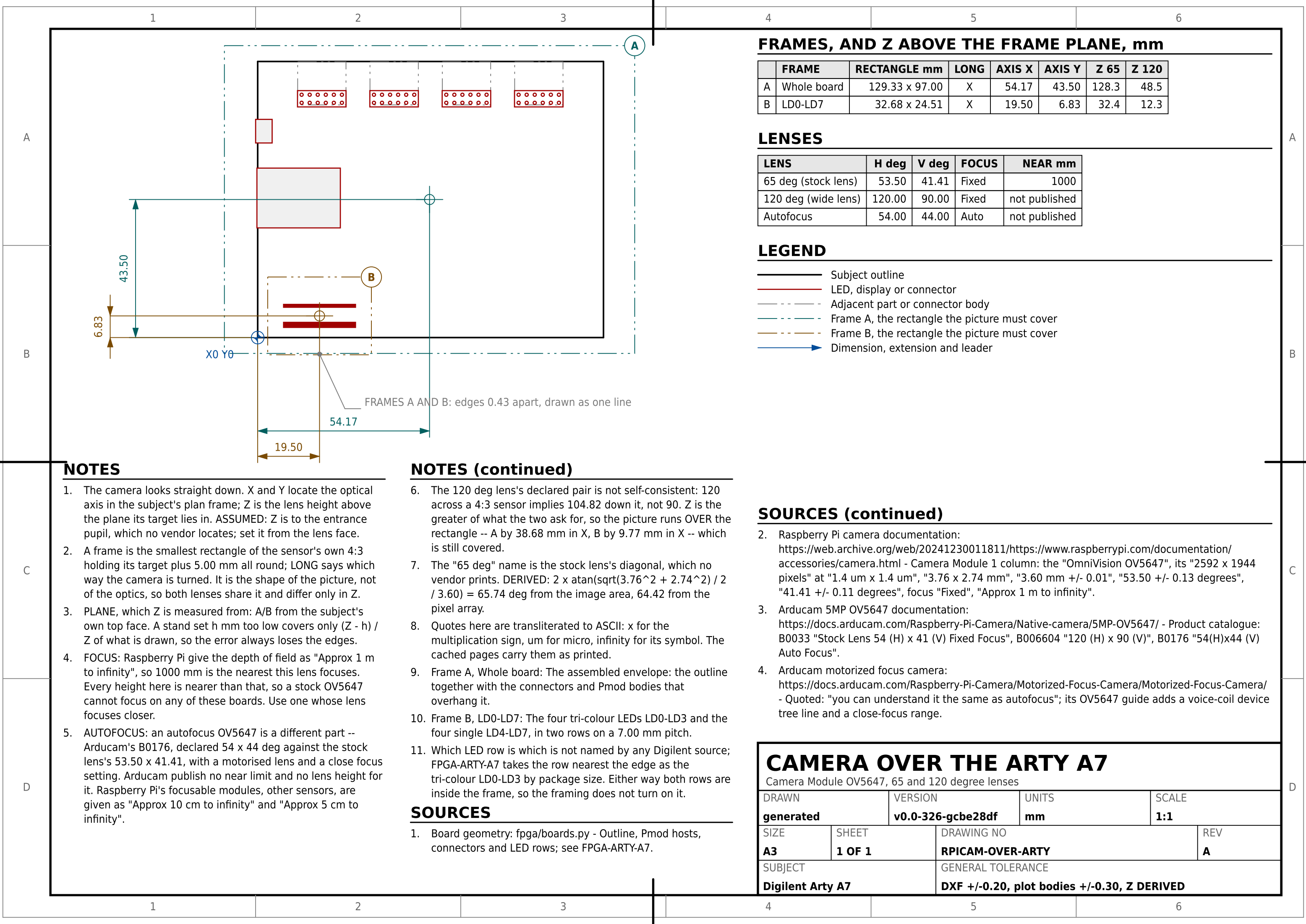
SOURCES

1. Plate geometry: [tinytapeout/mounting_plate/plate.py](#) - Outline and placements; see TT-MP-PLATE.
2. Indicator positions: [tinytapeout/boards.py](#) - LED and 7-segment footprints, in plate coordinates.
3. Raspberry Pi camera documentation: <https://web.archive.org/web/20241230011811/https://www.raspberrypi.com/documentation/accessories/camera.html> - Camera Module 1 column: the "OmniVision OV5647", its "2592 x 1944 pixels" at "1.4 um x 1.4 um", "3.76 x 2.74 mm", "3.60 mm +/- 0.01", "53.50 +/- 0.13 degrees", "41.41 +/- 0.11 degrees", focus "Fixed", "Approx 1 m to infinity".
4. Arducam 5MP OV5647 documentation: <https://docs.arducam.com/Raspberry-Pi-Camera/Native-camera/5MP-OV5647/> - Product catalogue: B0033 "Stock Lens 54 (H) x 41 (V) Fixed Focus", B006604 "120 (H) x 90 (V)", B0176 "54(H)x44 (V) Auto Focus".
5. Arducam motorized focus camera: <https://docs.arducam.com/Raspberry-Pi-Camera/Motorized-Focus-Camera/Motorized-Focus-Camera/> - Quoted: "you can understand it the same as autofocus"; its OV5647 guide adds a voice-coil device tree line and a close-focus range.

CAMERA OVER THE TT MOUNTING PLATE

Camera Module OV5647, 65 and 120 degree lenses

DRAWN generated		VERSION v0.0-326-gcbe28df		UNITS mm		SCALE 1:1	
SIZE A3	SHEET 1 OF 1		DRAWING NO RPICAM-OVER-PLATE				REV A
SUBJECT TT Mounting Plate			GENERAL TOLERANCE plate +/-0.20, LEDs +/-0.10, Z DERIVED				



FRAMES, AND Z ABOVE THE FRAME PLANE, mm

	FRAME	RECTANGLE mm	LONG	AXIS X	AXIS Y	Z 65	Z 120
A	Whole board	129.33 x 97.00	X	54.17	43.50	128.3	48.5
B	LD0-LD7	32.68 x 24.51	X	19.50	6.83	32.4	12.3

LENSES

LENS	H deg	V deg	FOCUS	NEAR mm
65 deg (stock lens)	53.50	41.41	Fixed	1000
120 deg (wide lens)	120.00	90.00	Fixed	not published
Autofocus	54.00	44.00	Auto	not published

LEGEND

- Subject outline
- LED, display or connector
- Adjacent part or connector body
- Frame A, the rectangle the picture must cover
- Frame B, the rectangle the picture must cover
- Dimension, extension and leader

NOTES

- The camera looks straight down. X and Y locate the optical axis in the subject's plan frame; Z is the lens height above the plane its target lies in. ASSUMED: Z is to the entrance pupil, which no vendor locates; set it from the lens face.
- A frame is the smallest rectangle of the sensor's own 4:3 holding its target plus 5.00 mm all round; LONG says which way the camera is turned. It is the shape of the picture, not of the optics, so both lenses share it and differ only in Z.
- PLANE, which Z is measured from: A/B from the subject's own top face. A stand set h mm too low covers only (Z - h) / Z of what is drawn, so the error always loses the edges.
- FOCUS: Raspberry Pi give the depth of field as "Approx 1 m to infinity", so 1000 mm is the nearest this lens focuses. Every height here is nearer than that, so a stock OV5647 cannot focus on any of these boards. Use one whose lens focuses closer.
- AUTOFOCUS: an autofocus OV5647 is a different part -- Arducam's B0176, declared 54 x 44 deg against the stock lens's 53.50 x 41.41, with a motorised lens and a close focus setting. Arducam publish no near limit and no lens height for it. Raspberry Pi's focusable modules, other sensors, are given as "Approx 10 cm to infinity" and "Approx 5 cm to infinity".

NOTES (continued)

- The 120 deg lens's declared pair is not self-consistent: 120 across a 4:3 sensor implies 104.82 down it, not 90. Z is the greater of what the two ask for, so the picture runs OVER the rectangle -- A by 38.68 mm in X, B by 9.77 mm in X -- which is still covered.
- The "65 deg" name is the stock lens's diagonal, which no vendor prints. DERIVED: $2 \times \text{atan}(\sqrt{3.76^2 + 2.74^2}) / 2 / 3.60 = 65.74$ deg from the image area, 64.42 from the pixel array.
- Quotes here are transliterated to ASCII: x for the multiplication sign, um for micro, infinity for its symbol. The cached pages carry them as printed.
- Frame A, Whole board: The assembled envelope: the outline together with the connectors and Pmod bodies that overhang it.
- Frame B, LD0-LD7: The four tri-colour LEDs LD0-LD3 and the four single LD4-LD7, in two rows on a 7.00 mm pitch.
- Which LED row is which is not named by any Digilent source; FPGA-ARTY-A7 takes the row nearest the edge as the tri-colour LD0-LD3 by package size. Either way both rows are inside the frame, so the framing does not turn on it.

SOURCES

- Board geometry: fpga/boards.py - Outline, Pmod hosts, connectors and LED rows; see FPGA-ARTY-A7.

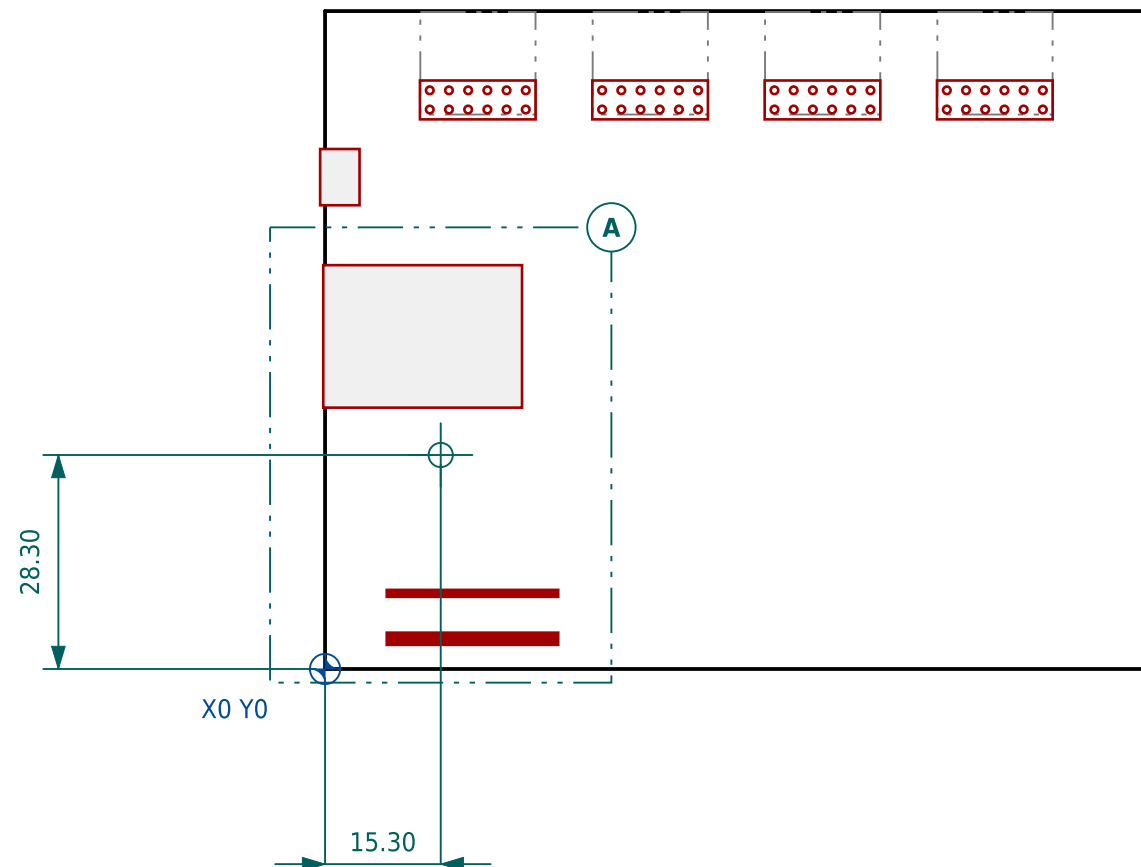
SOURCES (continued)

- Raspberry Pi camera documentation: <https://web.archive.org/web/20241230011811/https://www.raspberrypi.com/documentation/accessories/camera.html> - Camera Module 1 column: the "OmniVision OV5647", its "2592 x 1944 pixels" at "1.4 um x 1.4 um", "3.76 x 2.74 mm", "3.60 mm +/- 0.01", "53.50 +/- 0.13 degrees", "41.41 +/- 0.11 degrees", focus "Fixed", "Approx 1 m to infinity".
- Arducam 5MP OV5647 documentation: <https://docs.arducam.com/Raspberry-Pi-Camera/Native-camera/5MP-OV5647/> - Product catalogue: B0033 "Stock Lens 54 (H) x 41 (V) Fixed Focus", B006604 "120 (H) x 90 (V)", B0176 "54(H)x44 (V) Auto Focus".
- Arducam motorized focus camera: <https://docs.arducam.com/Raspberry-Pi-Camera/Motorized-Focus-Camera/Motorized-Focus-Camera/> - Quoted: "you can understand it the same as autofocus"; its OV5647 guide adds a voice-coil device tree line and a close-focus range.

CAMERA OVER THE ARTY A7

Camera Module OV5647, 65 and 120 degree lenses

DRAWN generated		VERSION v0.0-326-gcbe28df		UNITS mm		SCALE 1:1	
SIZE A3	SHEET 1 OF 1		DRAWING NO RPICAM-OVER-ARTY				REV A
SUBJECT Digilent Arty A7			GENERAL TOLERANCE DXF +/-0.20, plot bodies +/-0.30, Z DERIVED				



NOTES

1. The camera looks straight down. X and Y locate the optical axis in the subject's plan frame; Z is the lens height above the plane its target lies in. ASSUMED: Z is to the entrance pupil, which no vendor locates; set it from the lens face.
2. A frame is the smallest rectangle of the sensor's own 4:3 holding its target plus 5.00 mm all round; LONG says which way the camera is turned. It is the shape of the picture, not of the optics, so both lenses share it and differ only in Z.
3. PLANE, which Z is measured from: A from the subject's own top face. A stand set h mm too low covers only $(Z - h) / Z$ of what is drawn, so the error always loses the edges.
4. FOCUS: Raspberry Pi give the depth of field as "Approx 1 m to infinity", so 1000 mm is the nearest this lens focuses. Every height here is nearer than that, so a stock OV5647 cannot focus on any of these boards. Use one whose lens focuses closer.
5. AUTOFOCUS: an autofocus OV5647 is a different part -- Arducam's B0176, declared 54 x 44 deg against the stock lens's 53.50 x 41.41, with a motorised lens and a close focus setting. Arducam publish no near limit and no lens height for it. Raspberry Pi's focusable modules, other sensors, are given as "Approx 10 cm to infinity" and "Approx 5 cm to infinity".
6. The 120 deg lens's declared pair is not self-consistent: 120 across a 4:3 sensor implies 104.82 down it, not 90. Z is the greater of what the two ask for, so the picture runs OVER the rectangle -- A by 18.00 mm in Y -- which is still covered.

NOTES (continued)

7. The "65 deg" name is the stock lens's diagonal, which no vendor prints. DERIVED: $2 \times \text{atan}(\sqrt{3.76^2 + 2.74^2}) / 2 / 3.60 = 65.74$ deg from the image area, 64.42 from the pixel array.
8. Quotes here are transliterated to ASCII: x for the multiplication sign, um for micro, infinity for its symbol. The cached pages carry them as printed.
9. Frame A, LD0-LD7 + RJ45: The user LEDs together with the Ethernet jack body J9, which is where the link and activity LEDs are.
10. ASSUMED: the Ethernet LEDs are on the front face of the RJ45 jack, pointing out of the board edge, and cannot be seen from above at all. This frame covers the jack's BODY footprint, on the assumption that a light pipe adapter brings them to the top somewhere near it. Nothing here is a measurement of such an adapter.
11. FPGA-LP-ARTY draws one, and it puts its pipe tips 5.01 mm in FRONT of the jack's front face rather than over the body. That is inside this frame, which reaches further out again, but the frame is still the jack's body and the margin: treat its extent as provisional until an exit is dimensioned onto this sheet.

SOURCES

1. Board geometry: fpga/boards.py - Outline, Pmod hosts, connectors and LED rows; see FPGA-ARTY-A7.

FRAMES, AND Z ABOVE THE FRAME PLANE, mm

	FRAME	RECTANGLE mm	LONG	AXIS X	AXIS Y	Z 65	Z 120
A	LD0-LD7 + RJ45	45.15 x 60.21	Y	15.30	28.30	59.7	22.6

LENSES

LENS	H deg	V deg	FOCUS	NEAR mm
65 deg (stock lens)	53.50	41.41	Fixed	1000
120 deg (wide lens)	120.00	90.00	Fixed	not published
Autofocus	54.00	44.00	Auto	not published

LEGEND

- Subject outline
- LED, display or connector
- - - Adjacent part or connector body
- - - Frame A, the rectangle the picture must cover
- Dimension, extension and leader

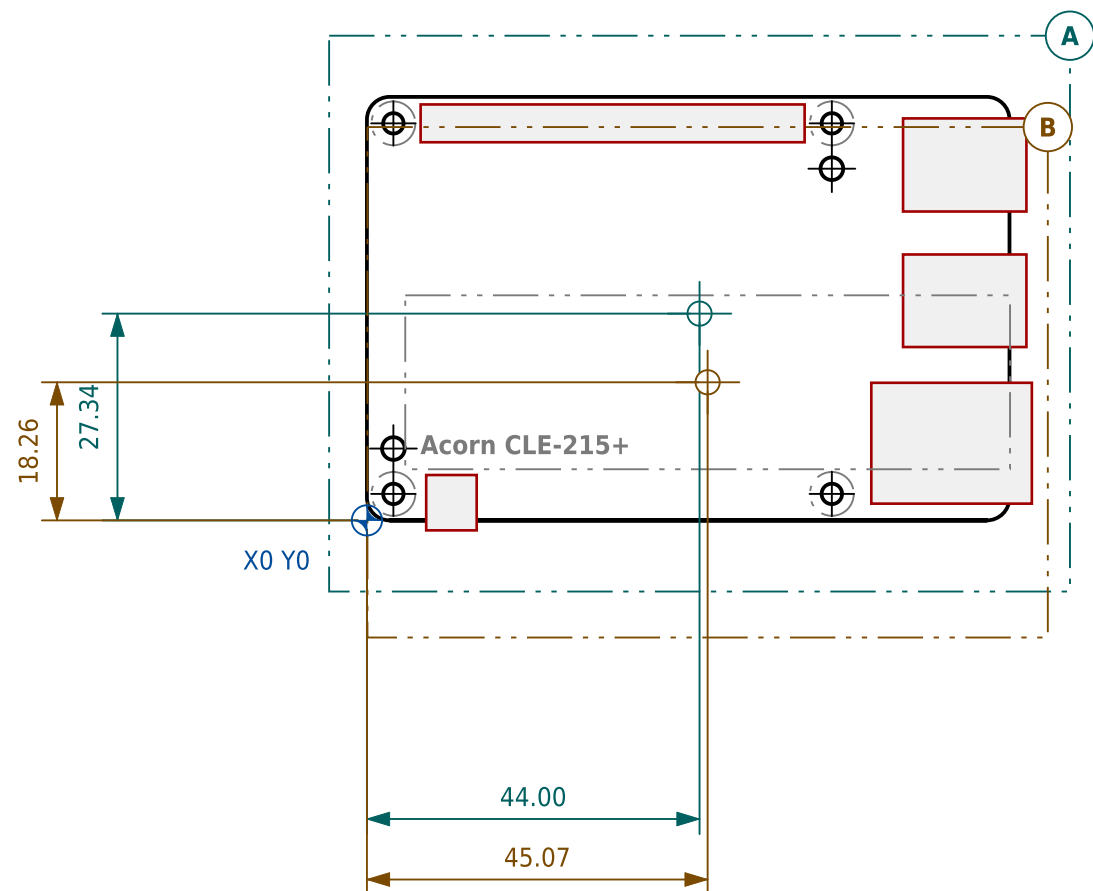
SOURCES (continued)

2. Raspberry Pi camera documentation: <https://web.archive.org/web/20241230011811/https://www.raspberrypi.com/documentation/accessories/camera.html> - Camera Module 1 column: the "OmniVision OV5647", its "2592 x 1944 pixels" at "1.4 um x 1.4 um", "3.76 x 2.74 mm", "3.60 mm +/- 0.01", "53.50 +/- 0.13 degrees", "41.41 +/- 0.11 degrees", focus "Fixed", "Approx 1 m to infinity".
3. Arducam 5MP OV5647 documentation: <https://docs.arducam.com/Raspberry-Pi-Camera/Native-camera/5MP-OV5647/> - Product catalogue: B0033 "Stock Lens 54 (H) x 41 (V) Fixed Focus", B006604 "120 (H) x 90 (V)", B0176 "54(H)x44 (V) Auto Focus".
4. Arducam motorized focus camera: <https://docs.arducam.com/Raspberry-Pi-Camera/Motorized-Focus-Camera/Motorized-Focus-Camera/> - Quoted: "you can understand it the same as autofocus"; its OV5647 guide adds a voice-coil device tree line and a close-focus range.

CAMERA OVER THE ARTY A7, ETHERNET LEDS INCLUDED

Camera Module OV5647, 65 and 120 degree lenses

DRAWN generated		VERSION v0.0-326-gcbe28df		UNITS mm		SCALE 1:1	
SIZE A3	SHEET 1 OF 1		DRAWING NO RPICAM-OVER-ETH				REV A
SUBJECT Diligent Arty A7			GENERAL TOLERANCE jack body +/-0.30, pipe exit ASSUMED, Z DERIVED				



NOTES

1. The camera looks straight down. X and Y locate the optical axis in the subject's plan frame; Z is the lens height above the plane its target lies in. ASSUMED: Z is to the entrance pupil, which no vendor locates; set it from the lens face.
2. A frame is the smallest rectangle of the sensor's own 4:3 holding its target plus 5.00 mm all round; LONG says which way the camera is turned. It is the shape of the picture, not of the optics, so both lenses share it and differ only in Z.
3. PLANE, which Z is measured from: A/B from the ACORN CARD's top face, not the Pi's -- nobody publishes how far the card stands above the Pi, so measure the stack. Set from the Pi's face the camera sits too low, and the picture at the card loses the ends of it. A stand set h mm too low covers only $(Z - h) / Z$ of what is drawn, so the error always loses the edges.
4. FOCUS: Raspberry Pi give the depth of field as "Approx 1 m to infinity", so 1000 mm is the nearest this lens focuses. Every height here is nearer than that, so a stock OV5647 cannot focus on any of these boards. Use one whose lens focuses closer.
5. AUTOFOCUS: an autofocus OV5647 is a different part -- Arducam's B0176, declared 54 x 44 deg against the stock lens's 53.50 x 41.41, with a motorised lens and a close focus setting. Arducam publish no near limit and no lens height for it. Raspberry Pi's focusable modules, other sensors, are given as "Approx 10 cm to infinity" and "Approx 5 cm to infinity".

NOTES (continued)

6. The 120 deg lens's declared pair is not self-consistent: 120 across a 4:3 sensor implies 104.82 down it, not 90. Z is the greater of what the two ask for, so the picture runs OVER the rectangle -- A by 29.31 mm in X, B by 26.91 mm in X -- which is still covered.
7. The "65 deg" name is the stock lens's diagonal, which no vendor prints. DERIVED: $2 \times \arctan(\sqrt{3.76^2 + 2.74^2}) / 3.60 = 65.74$ deg from the image area, 64.42 from the pixel array.
8. Quotes here are transliterated to ASCII: x for the multiplication sign, um for micro, infinity for its symbol. The cached pages carry them as printed.
9. Frame A, Whole assembly: Pi 5 with a Waveshare PoE M.2 HAT+ (B) and the Acorn seated in it; the HAT is the Pi's own 85 x 56 mm and its 2280 standoff reaches 88.00.
10. Frame B, Acorn card: 23 x 80 mm: the M.2 specification's Type 2280 outline with SQRL's own extra millimetre of width.
11. The Acorn's own LED positions are not published: SQRL issued no mechanical drawing and their site is gone, so frame B is the card, not its indicators. Its seated position and the assembly's far edge are accessories/parts.py's, which ACC-HAT-M2POE is drawn from too.

SOURCES

1. Board geometry: raspberry_pi/boards.py - Pi 5 outline and connectors; see RPI-5.
2. PCI Express M.2 Specification: PCI-SIG, Revision 1.0, 1 November 2013 - Figure 13: Type 2280 is 22 x 80, both +/-0.15.

FRAMES, AND Z ABOVE THE FRAME PLANE, mm

	FRAME	RECTANGLE mm	LONG	AXIS X	AXIS Y	Z 65	Z 120
A	Whole assembly	98.00 x 73.50	X	44.00	27.34	97.2	36.8
B	Acorn card	90.00 x 67.50	X	45.07	18.26	89.3	33.8

LENSES

LENS	H deg	V deg	FOCUS	NEAR mm
65 deg (stock lens)	53.50	41.41	Fixed	1000
120 deg (wide lens)	120.00	90.00	Fixed	not published
Autofocus	54.00	44.00	Auto	not published

LEGEND

- Subject outline
- LED, display or connector
- - - Adjacent part or connector body
- - - Frame A, the rectangle the picture must cover
- - - Frame B, the rectangle the picture must cover
- Dimension, extension and leader

SOURCES (continued)

3. SQRL Acorn CLE-215+ product page: <https://web.archive.org/web/2020/http://www.squirrelsresearch.com/acorn-cle-215-plus/> - Captured 2020; the site is gone. Quoted: "it is one millimeter wider than the official specifications."
4. Waveshare PoE M.2 HAT+ (B) dimension drawing: <https://www.waveshare.com/w/upload/d/d9/PoE-M.2-HAT-Plus-B-details-size.jpg> - Annotated 85.00, 56.00 and 3.00, "Unit: mm": the 3.00 is the standoff past the board edge. It dimensions no height.
5. Raspberry Pi camera documentation: <https://web.archive.org/web/20241230011811/https://www.raspberrypi.com/documentation/accessories/camera.html> - Camera Module 1 column: the "OmniVision OV5647", its "2592 x 1944 pixels" at "1.4 um x 1.4 um", "3.76 x 2.74 mm", "3.60 mm +/- 0.01", "53.50 +/- 0.13 degrees", "41.41 +/- 0.11 degrees", focus "Fixed", "Approx 1 m to infinity".
6. Arducam 5MP OV5647 documentation: <https://docs.arducam.com/Raspberry-Pi-Camera/Native-camera/5MP-OV5647/> - Product catalogue: B0033 "Stock Lens 54 (H) x 41 (V) Fixed Focus", B006604 "120 (H) x 90 (V)", B0176 "54(H)x44 (V) Auto Focus".
7. Arducam motorized focus camera: <https://docs.arducam.com/Raspberry-Pi-Camera/Motorized-Focus-Camera/Motorized-Focus-Camera/> - Quoted: "you can understand it the same as autofocus"; its OV5647 guide adds a voice-coil device tree line and a close-focus range.

CAMERA OVER THE ACORN CLE-215+

Camera Module OV5647, 65 and 120 degree lenses

DRAWN generated		VERSION v0.0-326-gcbe28df		UNITS mm		SCALE 1:1	
SIZE A3	SHEET 1 OF 1		DRAWING NO RPICAM-OVER-ACORN				REV A
SUBJECT Acorn CLE-215+, Pi 5			GENERAL TOLERANCE Pi 5 +/-0.20, card +/-0.20 DERIVED, Z DERIVED				